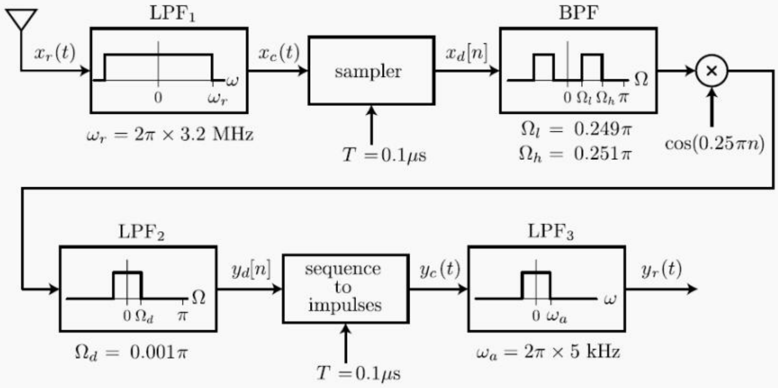
Problem 1：

Commercial AM radio stations broadcast radio frequencies within a limited range: $2\pi(f_c - 5\text{ kHz}) < \omega < 2\pi(f_c + 5\text{ kHz})$, where $f_c = \omega_c/(2\pi) = n \times 10$ kHz and n is an integer between 54 and 160. The system shown below is intended to decode one of the AM radio signals using DT signal processing methods. Assume that all of the filters are ideal.



sampler: $x_d[n] = x_c(nT)$

sequence to impulses: $y_c(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT)$

Part a. Determine the center frequency f_c for the AM station that this receiver will detect.

Part b. Which of the following statement(s) is/are correct?

b1. Increasing the cutoff frequency ω_r of LPF$_1$ by a factor of 1.5 will cause aliasing.

b2. Decreasing the cutoff frequency ω_r of LPF$_1$ by a factor of 2 will have no effect on the output $y_r(t)$.

b3. Halving the sampling interval T would have no effect on the output $y_r(t)$.

b4. Doubling the sampling interval T would have no effect on the output $y_r(t)$.

Part c. Which of the following statement(s) is/are correct?

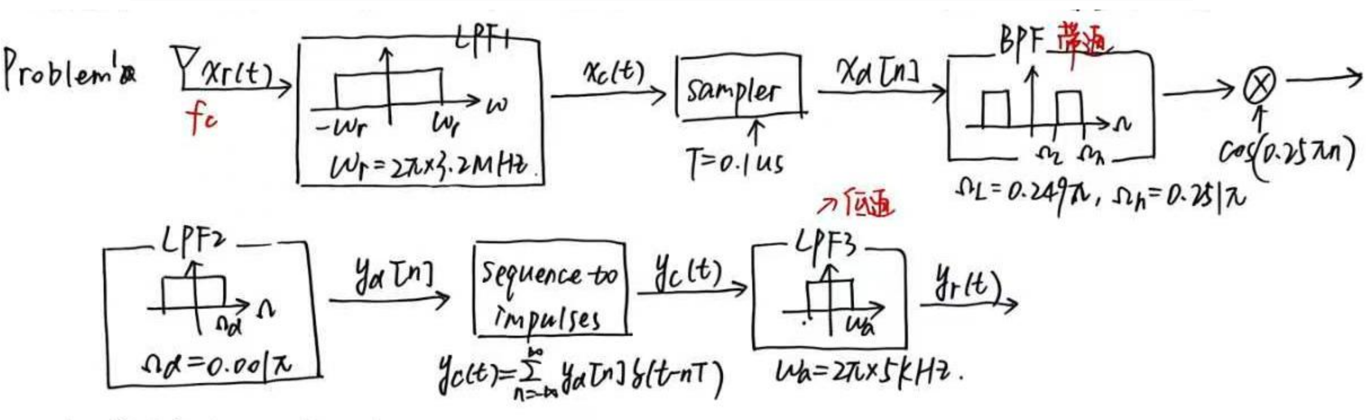
c1. Increasing the cutoff frequency Ω_d of LPF$_2$ will change $y_r(t)$ by adding signals from unwanted radio stations.

c2. Increasing the cutoff frequency Ω_d of LPF$_2$ will change $y_r(t)$ because aliasing will occur.

c3. Doubling the cutoff frequency Ω_d of LPF$_2$ will have no effect on $y_r(t)$.

c4. Halving the cutoff frequency Ω_d of LPF$_2$ will have no effect on $y_r(t)$.

Problem1:



(a) 采样频率 $f_s = \frac{1}{T} = \frac{1}{0.1us} = 10MHz$.

$x(t) \Rightarrow x[n]$

$\omega = 2\pi f_c$ $\quad \Omega = 2\pi f_c T = 2\pi \frac{f_c}{f_s}$

对BPF: $\Omega_c = 0.25\pi$. 就是 $x_d[n]$ 的角频率.

$\Rightarrow \Omega_c = 2\pi \frac{f_c}{f_s} \Rightarrow f_c = 1250kHz$.

(b) b1. 增加LPF1的截止频率 ω_r 1.5倍将导致混叠. X

$f_r=3.2MHz$, $f_r'=4.8MHz$. $f_r'<\frac{1}{2}f_s=5MHz$ ⇒ 不会混叠.

b2. 降低LPF2的截止频率 ω_r 2倍, 对输出 $y_r(t)$ 无影响. ✓

$f_r''=1.6MHz$, ~~$f_r''\in(f_c-5kHz, f_c+5kHz)$~~ $f_r''>f_c+5kHz$.

b3. 将T减半, 对 $y_r(t)$ 无影响. X.

$\Omega_c=$ ~~0.5~~ 0.125π 过不了BPF.

b4. 将T加倍, 对 $y_r(t)$ 无影响 X.

(c) c1. 增加LPF2的 Ω_α, 将通过加入不需要的其他信号来改变 $y_r(t)$ X

在BPF已滤除.

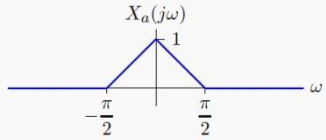
c2. 将LPF2的 Ω_α 增加, ⇒ 改变 $y_r(t)$ ⇒ 混叠. X

c3. Ω_α 加倍, 对 $y_r(t)$ 无影响. ✓.

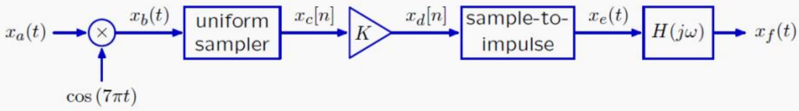
~~C4~~ C4. Ω_α 减半, 对 $y_r(t)$ 无影响 X.

Problem 2：

The Fourier transform of a signal $x_a(t)$ is given below.



This signal passes through the following system



where $x_c[n] = x_b(nT)$ and

$$x_e(t) = \sum_{n=-\infty}^{\infty} x_d[n]\delta(t - nT)$$

and

$$H(j\omega) = \begin{cases} T & \text{if } |\omega| < \frac{\pi}{T} \\ 0 & \text{otherwise.} \end{cases}$$

a. Sketch the Fourier transform of $x_f(t)$ for the case when $K = 1$ and $T = 1$.

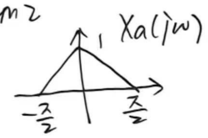
Use your sketch to determine an expression for $X_f(j\omega)$ for the following intervals:

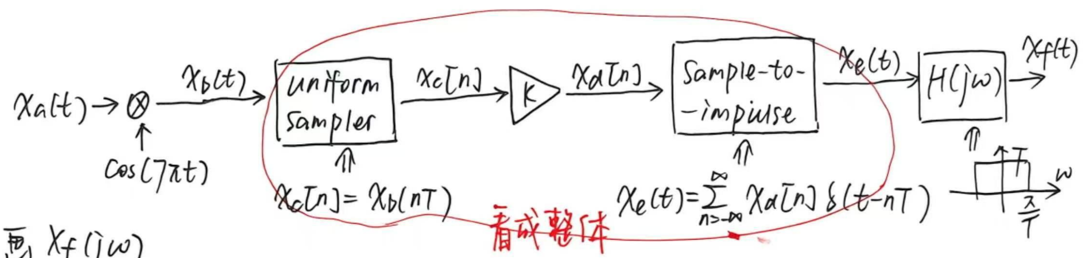
$0 < \omega < \pi/2$:	
$\pi/2 < \omega < \pi$:	
$\pi < \omega < 3\pi/2$:	
$3\pi/2 < \omega < 2\pi$:	

b. Is it possible to adjust T and K so that $x_f(t) = x_a(t)$?

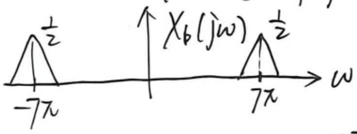
If yes, specify a value T and the corresponding value of K (there may be multiple solutions, you need only specify one of them). If no, write **none**.

Problem 2





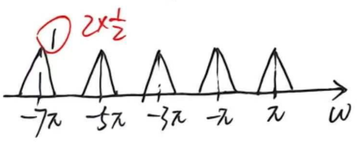
(a) 对 $K=1, T=1$, 画 $X_f(j\omega)$



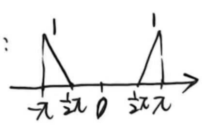
$$X_e(j\omega) = \frac{K}{T}\sum_{n=-\infty}^{\infty} X_b(j(\omega - n\omega_s))$$

$K=1, T=1$ 时, $\omega_s = \frac{2\pi}{T} = 2\pi\ rad/s$, $X_e(j\omega) = \sum_{n=-\infty}^{\infty} X_b(j(\omega - 2\pi n))$

⇒ $X_e(j\omega)$:



⇒ $X_f(j\omega)$:



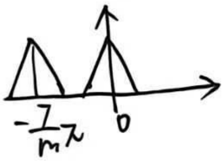
则 $X_f(j\omega)=\begin{cases}0, & 0<\omega<\frac{\pi}{2}\\ \frac{2}{\pi}\omega-1, & \frac{\pi}{2}<\omega<\pi \\ 0, & \pi<\omega<2\pi\end{cases}$

(b) 调整 K、T，使 $x_f(t)=x_a(t)$.

要让 $X_e(j\omega)$有 

的形状. $\Rightarrow \omega_s=\frac{7\pi}{m}$.

$\omega_s=\frac{2\pi}{T} \Rightarrow T=\frac{2m}{7}$.



$\begin{cases} -\frac{7}{m}\pi+\frac{1}{2}\pi<-\frac{1}{2}\pi \\ -\frac{7}{m}\pi+\frac{1}{2}\pi<-\frac{\pi}{T}<-\frac{1}{2}\pi\end{cases} \Rightarrow m\in[1,7]$

$X_e(j\omega)$的峰值: $\frac{K}{T}\cdot 2\cdot\frac{1}{2}=\frac{K}{T}$.

$X_f(j\omega)$的峰值: $\frac{K}{T}\cdot T=K \Rightarrow K=1$.

$\Rightarrow \begin{cases} K=1 \\ T=\frac{2m}{7}, \quad m=1,2,3,4,5,6,7.\end{cases}$

Problem 3

8.8. Consider the modulation system shown in Figure P8.8. The input signal $x(t)$ has a Fourier transform $X(j\omega)$ that is zero for $|\omega| > \omega_M$. Assuming that $\omega_c > \omega_M$, answer the following questions:

(a) Is $y(t)$ guaranteed to be real if $x(t)$ is real?

(b) Can $x(t)$ be recovered from $y(t)$?

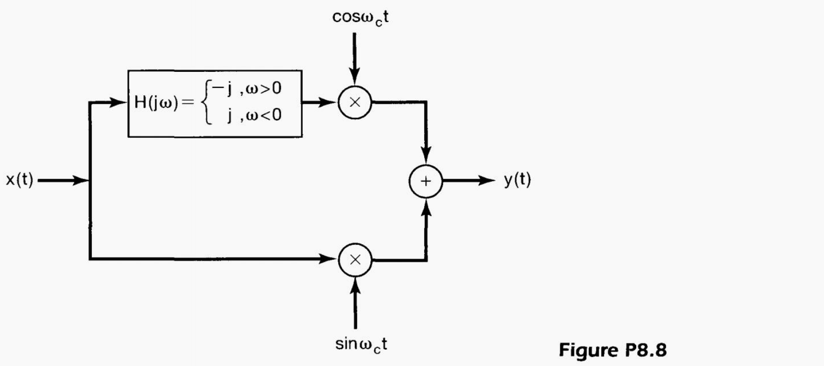
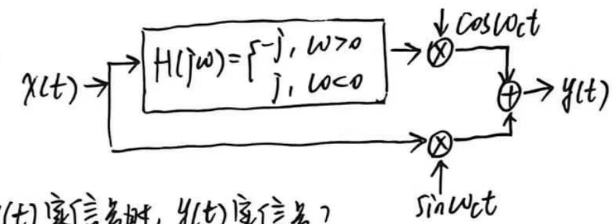


Figure P8.8

Problem3.



(a) $x(t)$实信号时，$y(t)$实信号？

$h(t)$是否实信号 → $h(j\omega)=h^*(-j\omega)$ $\begin{cases} \omega>0\text{时}, h(j\omega)=-j,\ h(-j\omega)=j,\ \text{共轭}. \\ \omega<0\text{时}, h(j\omega)=j,\ h(-j\omega)=-j,\ \text{共轭} \end{cases}$

⇒ $h(t)$是实信号

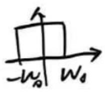
⇒ $y(t)=\cos\omega_c t\cdot(x(t)*h(t))+\sin\omega_c t\cdot x(t)$ 实信号.

(b) 令 $x(t)*h(t)=x_1(t)$.

不用纠结$x_1(t)$是什么 → 高频滤除 $x_1(t)$ → ×sin 不是cos (保留$x(t)$).

$y(t)=\cos\omega_c t\cdot x_1(t)+\sin\omega_c t\cdot x_2(t)$

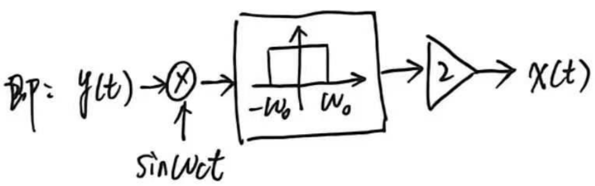
$y(t)\cdot\sin\omega_c t=\underbrace{\tfrac{1}{2}x_1(t)\cdot\sin(2\omega_c t)}_{\text{在}2\omega_c\text{附近}}+\tfrac{1}{2}x(t)-\tfrac{1}{2}x(t)\cos(2\omega_c t)$

过LPF， 使 $\omega_M<\omega_0<2\omega_c-\omega_M$

$y'(t)=\tfrac{1}{2}x(t)$

↓×2

$x(t)$

即：

Problem 4

8.21. In Sections 8.1 and 8.2, we analyzed the sinusoidal amplitude modulation and demodulation system of Figure 8.8, assuming that the phase θ_c of the carrier signal was zero.

(a) For the more general case of arbitrary phase θ_c in the figure, show that the signal in the demodulation system can be expressed as

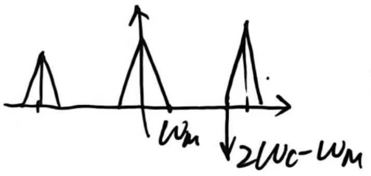
$$w(t) = \frac{1}{2}x(t) + \frac{1}{2}x(t)\cos(2\omega_c t + 2\theta_c).$$

(b) If $x(t)$ has a spectrum that is zero for $|\omega| \geq \omega_M$, determine the relationships required among w_{co} [the cutoff frequency of the ideal lowpass filter in Figure 8.8(b)], ω_c (the carrier frequency), and ω_M so that the output of the lowpass filter is proportional to $x(t)$. Does your answer depend on the carrier phase θ_c?

Problem 4. (a) $w(t) = x(t)\cos^2(\omega_c t + \theta_c)$

$= \frac{1}{2}x(t) + \frac{1}{2}x(t)\cos(2\omega_c t + 2\theta_c)$

(b) $w(t) = \frac{1}{2}x(t) + \frac{1}{2}x(t)\cdot\cos(2\omega_c t)\cos(2\theta_c) + \frac{1}{2}x(t)\sin(\omega_c t)\sin(2\theta_c)$



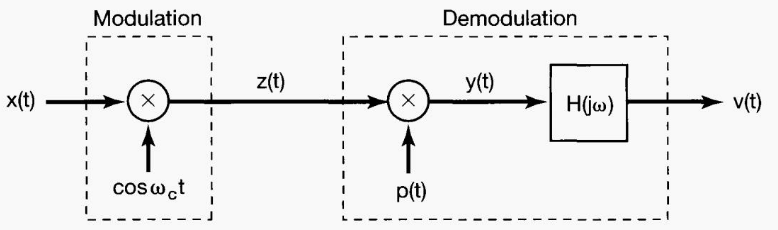
$$\begin{cases} \omega_M < \omega_{co} < 2\omega_c - \omega_M \\ \omega_M < 2\omega_c - \omega_M \Rightarrow \omega_c > \omega_M \end{cases}$$ 与θ_c无关.

Problem 5

8.35. The modulation-demodulation scheme proposed in this problem is similar to sinusoidal amplitude modulation, except that the demodulation is done with a square wave with the same zero-crossings as $\cos\omega_c t$. The system is shown in Figure P8.35(a); the relation between $\cos\omega_c t$ and $p(t)$ is shown in Figure P8.35(b). Let the input signal $x(t)$ be a band-limited signal with maximum frequency $\omega_M < \omega_c$, as shown in Figure P8.35(c).

(a) Sketch and dimension the real and imaginary parts of $Z(j\omega)$, $P(j\omega)$, and $Y(j\omega)$, the Fourier transforms of $z(t)$, $p(t)$, and $y(t)$, respectively.

(b) Sketch and dimension a filter $H(j\omega)$ so that $v(t) = x(t)$.



(a)

Figure P8.35

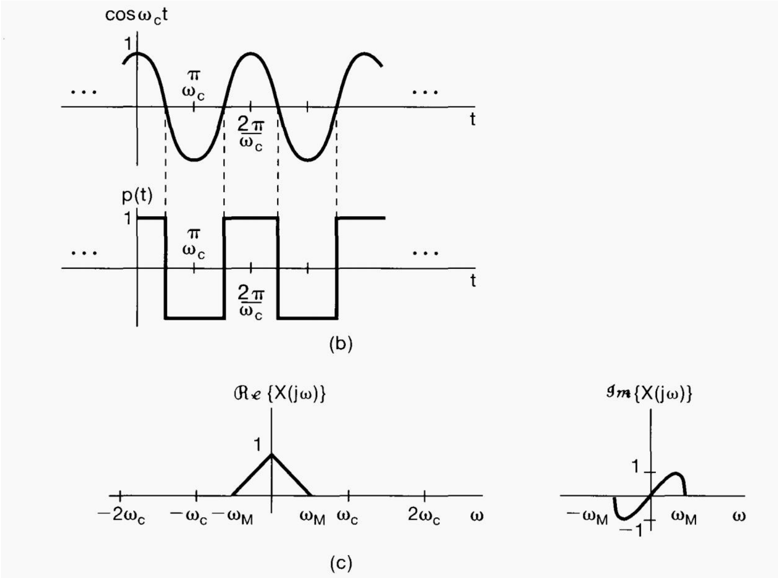
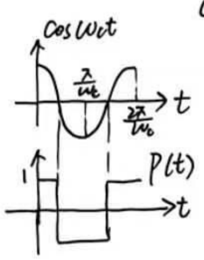


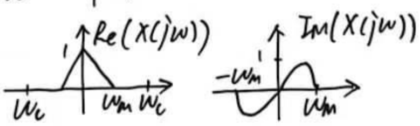
Figure P8.35 *Continued*

Problem 5

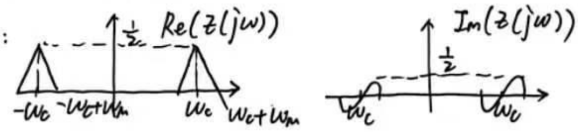
$x(t) \to \otimes \xrightarrow{z(t)} \otimes \xrightarrow{y(t)} \boxed{H(j\omega)} \to v(t)$

$\cos\omega_c t$ $\qquad$ $p(t)$





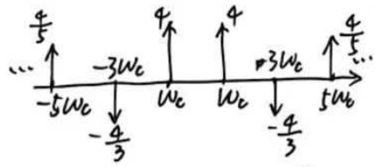
(a) 对 $Z(j\omega)$:



对 $P(j\omega)$: 对  $a_{k1} = \frac{\sin k\omega_0 \cdot \frac{\pi}{2\omega_c}}{k\pi} = \frac{\sin\frac{k\pi}{2}}{k\pi}$

对 $p(t)$, $a_{k2} = 2a_{k1}$, $a_{02} = 2a_{01} - 1 = 0$.

$\Rightarrow P(j\omega) = \sum_{k=-\infty}^{\infty} a_k[2\pi\delta(\omega - k\omega_c)]$

 $P(j\omega)$实部. 虚部全为0.

对 $Y(j\omega)$: $y(t) = z(t)\cdot p(t) \Rightarrow Y(j\omega) = \frac{1}{2\pi} Z(j\omega) * P(j\omega)$.

② $\Rightarrow y(t) = x(t)\cdot\cos\omega_c t\cdot p(t) = x(t)\cdot|\cos\omega_c t|$

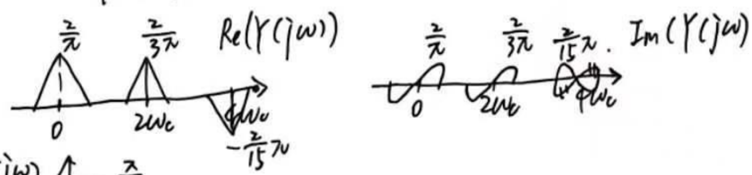
① $Y(j\omega) = \frac{1}{2\pi} Z(j\omega) * \sum_{k=-\infty}^{\infty} a_k[2\pi\delta(\omega - k\omega_c)]$

$= \int_{-\infty}^{\infty}\left[\sum_{k=-\infty}^{\infty} a_k\delta(\theta - k\omega_c)\right] Z(j(\omega-\theta))\,d\theta$ $\qquad$ $\int_{-\infty}^{\infty} f(x)\delta(x-x_0)\,dx = f(x_0)$

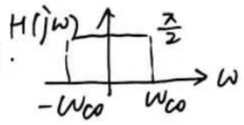
$= \sum_{k=-\infty}^{\infty} a_k\left[\int_{-\infty}^{\infty} \delta(\theta - k\omega_c) Z(j(\omega-\theta))\,d\theta\right]$

$= \sum_{k=-\infty}^{\infty} a_k \cdot Z(j(\omega - k\omega_c))$

落在偶数倍上.



(b).



$\omega_M < \omega_{c0} < 2\omega_c - \omega_M$.

Problem 6

8.36. The accurate demultiplexing-demodulation of radio and television signals is generally performed using a system called the superheterodyne receiver, which is equivalent to a tunable filter. The basic system is shown in Figure P8.36(a).

(a) The input signal $y(t)$ consists of the superposition of many amplitude-modulated signals that have been multiplexed using frequency-division multiplexing, so that each signal occupies a different frequency channel. Let us consider one such channel that contains the amplitude-modulated signal $y_1(t) = x_1(t)\cos\omega_c t$, with spectrum $Y_1(j\omega)$ as depicted at the top of Figure P8.36(b). We want to demultiplex and demodulate $y_1(t)$ to recover the modulating signal $x_1(t)$, using the system of Figure P8.36(a). The coarse tunable filter has the spectrum $H_1(j\omega)$ shown at the bottom of Figure P8.36(b). Determine the spectrum $Z(j\omega)$ of the input signal to the fixed selective filter $H_2(j\omega)$. Sketch and label $Z(j\omega)$ for $\omega > 0$.

(b) The fixed frequency-selective filter is a bandpass type centered around the fixed frequency ω_f, as shown in Figure P8.36(c). We would like the output of the filter with spectrum $H_2(j\omega)$ to be $r(t) = x_1(t)\cos\omega_f t$. In terms of ω_c and ω_M, what constraint must ω_T satisfy to guarantee that an undistorted spectrum of $x_1(t)$ is centered around $\omega = \omega_f$?

(c) What must G, α, and β be in Figure P8.36(c) so that $r(t) = x_1(t)\cos\omega_f t$?

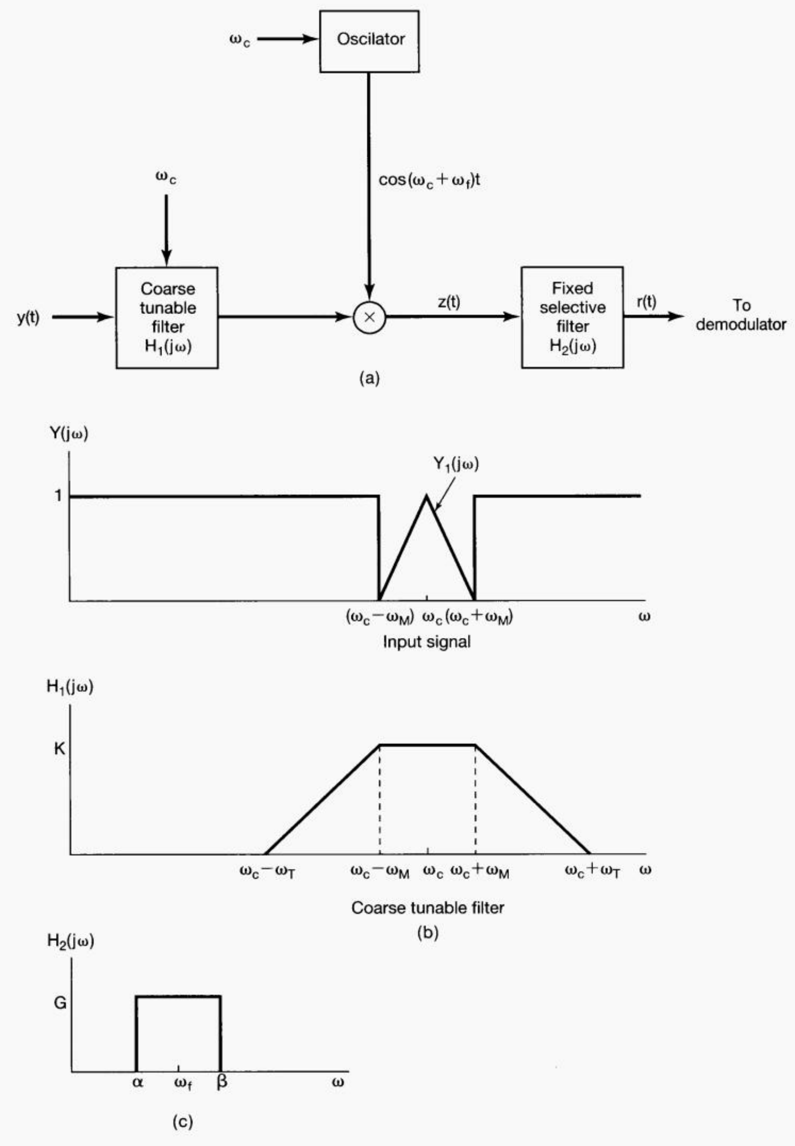
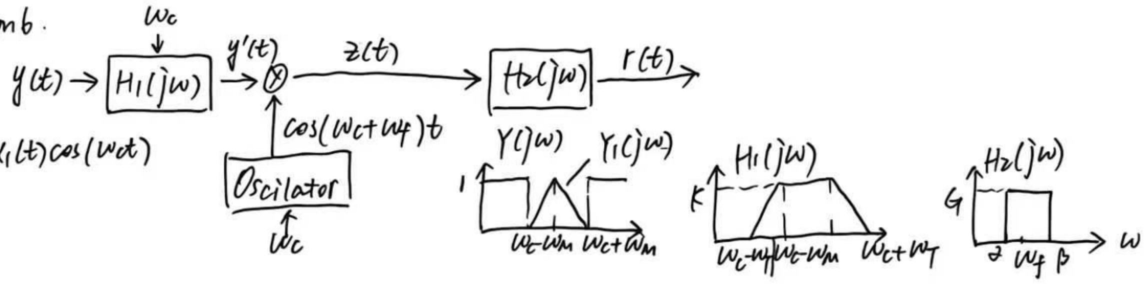


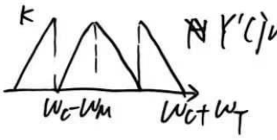
Figure P8.36

Problem6.

$y_1(t) = x_1(t)\cos(w_c t)$



(a)



对 $Y'(jw)$ 左右移 (w_c+w_4) $\Rightarrow$ $Z(jw)$

(-w_c 右移 w_c+w_4 得到)(都是关于y对称)

(b) 希望 $r(t) = x_1(t)\cos(w_4 t)$. 则 w_T 要满足什么条件?, 保证 $x_1(t)$ 无失真频谱以 $w=w_4$ 为中心?

只有原始目标. 不混杂其他.

$y'(t)$ ~~z(t)~~ 是否存在另一频率 w_0, 变成 $z(t)$ 后会被 $H_2(jw)$ 保留.

$|w_0 - (w_c+w_4)| = w_4 \Rightarrow w_0 = w_c + 2w_4$

其实-w右移和w左移是一样效果. ∴只考虑一种

$y'(t)$ 不能存在 $w_0 \Rightarrow$ w_0 被 $H_1(jw)$ 滤除.

$w_c + w_T \le w_c + 2w_4 - w_M \Rightarrow w_T \le 2w_4 - w_M$

(c) 求 G、α、β, 使 $r(t) = x_1(t)\cos(w_4 t)$.

(一定是轴对称的)

$\alpha = w_c - w_M$, $\beta = w_c + w_M$.

$\frac{1}{2}K \cdot G = 1 \Rightarrow G = \frac{2}{K}$

相当于

(默认 $x_1(t)$ 是基带信号 $\Rightarrow$ 中心频率是0)